# A completely specified Diophantine consistency equation

Fifteen strictly positive integer unknowns; all fixed constants defined on this page.

The unknowns are

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma \in \mathbb{Z}_{>0}.$$

Use the following polynomial abbreviations (no extra unknowns):

$$\begin{gathered} q = \beta J + 1, \quad C = q - F - \alpha, \quad X = wq, \\ Y = sq^3, \quad E = XY, \quad k = \eta + \zeta, \quad c = kY + \eta, \\ a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\ D = X + ac + \gamma H, \\ R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \\ V = c(Tf - 1) - Rf^2. \end{gathered}$$

The equation is

$$\boxed{N_0 N_1 N_a N_k N_t N_s - 1 = 0,}$$

where

$$\begin{aligned} N_0 &= \tau^2 - (XY^2)(XY^2 + 1)k^2, \\ N_1 &= D^2 - \Delta c^2, \\ N_a &= \Delta(f^2 - 1)(V^2 - y^2) + y^2, \\ N_k &= k - hE - R, \\ N_t &= (K + w)C + (q - F) - z(q - 1), \\ N_s &= f^2 - \Delta(ic^2)^2. \end{aligned}$$

**Fixed coefficients.** All sums below have the displayed fixed finite limits. Division and floor are used only to define constants; the equation above is polynomial. Put

$$\begin{gathered} \rho = 2^{89}, \quad B = \rho^{4831}, \quad \beta = B - 1, \\ \lambda = 2^{29}, \quad \theta = 2^{42}, \quad Q = 2^{40}, \end{gathered}$$

$$G_m(x) = \sum\nolimits_{j=0}^{m-1} x^j,$$

$$\mu = 510 + 2^{17} + 2^{26} + 2^{28} + \lambda(2^{44} - 1).$$

Let

$$\begin{aligned} \mathcal{S} = {} & \{0, \dots, 456\} \cup \{495, \dots, 504\} \cup \{507, 508\} \\ & \cup \{517, \dots, 537\} \cup \{539, \dots, 549\} \\ & \cup \{567, \dots, 583\} \cup \{1203, 1807\}. \end{aligned}$$

Then

$$\begin{aligned} M_C &= B - 1 - \sum\nolimits_{e \in \mathcal{S} \setminus \{1\}} \rho^e - 2\rho^{507}, \\ M_F &= B - 1 + 4 + \mu(\rho^{2409} + \rho^{3015}) \\ &\quad + \rho^{528} G_{10}(\rho) + \rho^{585} G_{11}(\rho) + \rho^{1203} + \rho^{1807} \\ &\quad + (2^{87} - 1)\rho^{4823} G_4(\rho) + (2^{38} - 1)\rho^{4827}, \\ K &= \rho^{33} + \rho^{68} + \rho^{664} + \rho^{1268} + \rho^{4877} \\ &\quad + (\rho^{2409} + \rho^{3015})\Phi(1/\rho), \end{aligned}$$

where the rational intermediate expression is

$$\begin{aligned} \Phi(x) = {} & O G_{457}(x) + \lambda \sum\nolimits_{i=0}^{456} \ell_i x^{r_i} \\ & - (255\, 2^9 + 2^{27}) x^{374} - (255\, 2^{18} + 2^{27}) x^4 \\ & + \lambda\big[(x^{539} + 2^{21} x^{517}) G_{11}(2x) \\ & \quad + (2^{11} x^{528} + 2^{32} x^{495}) G_{10}(2x) \\ & \quad + \theta(x^{539} + x^{1203} + x^{1807})\big] \\ & + \big[(2^9 + \lambda(2 + 2^{33}))x^{567} \\ & \quad + (2^{18} + \lambda(2^{22} + 2^{33}))x^{575}\big] G_8(2x) \\ & + (2^{27} + \lambda(2^{11} + 2^{32}))x^{583}. \end{aligned}$$

The resulting $\beta, K, M_C, M_F$ are fixed positive integers.

**The ranks $r_i$ and labels $\ell_i$.** First define the following 457 fixed quadruples $t_i = (w_i, n_i, e_i, s_i)$. From the integer $\mathcal{A}$ printed below, for $0 \le j < 430$ put

$$\begin{gathered} b_j = \left\lfloor \mathcal{A} / 1024^j \right\rfloor \bmod 1024, \quad u_j = b_j \bmod 256, \\ v_j = \lfloor b_j / 256 \rfloor \bmod 2, \quad d_j = \lfloor b_j / 512 \rfloor . \end{gathered}$$

Here $a \bmod b = a - b\lfloor a/b \rfloor$. Set

$$t_{2+j} = \big((1 - d_j)(512 + u_j),\ 512 + j,\ d_j(256 + u_j),\ v_j\big).$$

The remaining quadruples are

| index | range | $t_i$ |
|---|---|---|
| 0 | | $(2, 3, 3, 1)$ |
| 1 | | $(5, 0, 0, 0)$ |
| $432 + j$ | $0 \le j < 2$ | $(0, 942 + j, 0, 942 + j)$ |
| $434 + j$ | $0 \le j < 2$ | $(0, j, 0, j)$ |
| $436 + j$ | $0 \le j < 4$ | $(0, (j \bmod 2) + 942\lfloor j/2 \rfloor, 0, 6)$ |
| 440 | | $(2, 2, 2, 0)$ |
| 441 | | $(3, 4, 4, 698)$ |
| 442 | | $(4, 4, 4, 0)$ |
| $443 + j$ | $0 \le j < 3$ | $(1, 6, 1, 2 + j)$ |
| $446 + 3j + k$ | $0 \le j, k < 3$ | $(j^2, 5, k, 5)$ |
| 455 | | $(256, 0, 0, 512)$ |
| 456 | | $(0, 0, 512, 512)$ |

Define, for these fixed indices,

$$\begin{aligned} \kappa_i &= 2^{30} w_i + 2^{20} n_i + 2^{10} e_i + s_i, \\ \ell_i &= 2w_i + 2^{11} n_i + 2^{22} e_i + 2^{32} s_i, \\ r_0 &= 0, \quad r_1 = 1, \\ r_i &= 2 + \sum\nolimits_{j=2}^{456} \left\lfloor \frac{Q + \kappa_i - \kappa_j - 1}{Q} \right\rfloor \quad (2 \le i \le 456), \\ O &= 1 + 255\, 2^9 + 255\, 2^{18} + 2^{27} + \lambda. \end{aligned}$$

All subscripts in these definitions are bound constant indices, not additional Diophantine unknowns.

**The integer $\mathcal{A}$.** Digit values are $0, \dots, 61$ in this alphabet: `0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz`

For the following 722 characters, put $\mathcal{A} = \sum_{j=0}^{721} \epsilon_j 62^{721-j}$, where $\epsilon_j$ is the value of character $j$ (starting at zero).

```
RCtYkANMQ1gmPqxuKhLmpi1mjCvu243YjyAxv9hbC2UsurRLykLFhuHVBGGUZ5
whVShVOFQcVVQSR2GS6JAW1DGYxGKlA4rJavFCW5i63LLSs8xEGvWTbLjSFULV
ODe48Kw2ghEaa5Q9PZZVHtki7C45jfVYROOzvVT9DAGjLOBlBSHQU8WPUF3SpC
kTfDQF7mUboMEAeyPWARCyClpnxtqgsit318hQgiIdKauxc6memnYNtycBOs7z
K8dMgYaLRxNjlcLvJzZ9Z5wdCG3wjd5P6ZVqC2SXVaimqXrTY5dNxbCH2B5VKA
Sp16kht0JV65p4VclYsI6zi6nIr7elpUCOX7IHdu6qKiu6Kt5MqlYZjChEzG6y
W1WsPdMB7lPL3oOCFqFncCtVEMUounam1fyhduQYoTyTCIEMqcD2BPQfMVaBa9
BZHQ8k7rpS3eiM7Rsak5uUsoP1Tf1uhWshndLuN8yz7UdD5QXM6AJ3ovPdWsqO
HGBeRaR8kch4R9JVbkAxS2ByaeE2O6czIT0sWtpasWDdZB9FuDQUFYhSvPPPr8
nSQmnMkF90CBzWkpeP81azGOWtiT2XTxpaLcR4ecaK1uZyvBleYknk6QTwECl5
XJ3nJBoP4sMTGaV6unoGzq9s1KN4KtXrVRuTpi6EuqizW9njcp9Uikq1VNjVtw
W1AGLJu3eaWgOEEF7vROqLuNbzEheYnyi4SORWVp
```

Absence of positive integer solutions is equivalent to Con(ZFC) by the inherited native/Pell argument and the checked P69–P67 reduction. The full native calibration and its internal ZFC formalization remain open Lean obligations. This page specifies the equation, not its proof.